Lecture 13: Data I

> 第 13 讲：数据 I

Previous lectures: how to train a model *given data*

> 前几讲：给定数据，如何训练模型

Next two lectures: *what data* should we train on?

> 接下来两讲：我们应该在什么数据上训练？

Summary

> 总结

- Key lesson: Data does not fall from the sky. You have to work to get it.

> 关键启示：数据不会从天而降，你必须付出努力去获取。

- Live service → raw data → processed data (transformation, filtering, deduplication)

> 在线服务 → 原始数据 → 加工后的数据（转换、过滤、去重）

- Data is the key ingredient that differentiates language models

> 数据是让语言模型拉开差距的关键要素

- Legal and ethical issues (e.g., copyright and privacy)

> 法律与伦理问题（如版权和隐私）

- Much of this pipeline is heuristic, many opportunities to improve!

> 这条流水线的许多环节都是启发式的，有很大的改进空间！

Data is the most important thing to get right in training language models.

> 在训练语言模型时，把数据做对是最重要的事。

One justification: let's see what companies disclose.

> 一个佐证：看看各家公司都披露了什么。

Open-weight models (e.g., Llama 3 [Grattafiori+ 2024] have full transparency into architecture

> 开放权重的模型（如 Llama 3 [Grattafiori+ 2024]）在架构上完全透明

...and even training procedures

> ……甚至训练流程也透明

...but basically no information on data.

> ……但在数据方面基本没有任何信息。

3.1 Pre-Training Data

We create our dataset for language model pre-training from a variety of data sources containing knowledge until the end of 2023. We apply several de-duplication methods and data cleaning mechanisms on each data source to obtain high-quality tokens. We remove domains that contain large amounts of personally identifiable information (PII), and domains with known adult content.

Reasons for secrecy:

> 保密的原因：

1. Competitive dynamics

> 竞争因素

2. Copyright liability

> 版权责任

- Before foundation models, data work meant heavy annotation of labeled data for supervised learning.

> 在基础模型出现之前，数据工作意味着为监督学习对数据进行繁重标注。

- Now there's less annotation, but there's still a lot of curation and cleaning.

> 现在标注少了，但仍有大量的筛选与清洗工作。

- Data is fundamentally a long-tail problem, scales with human effort (unlike architectures, systems).

> 数据本质上是一个长尾问题，规模随人力投入增长（与架构、系统不同）。

Stages of training:

> 训练阶段：

1. Pre-training: train on raw text (e.g., documents from the web)

> 预训练：在原始文本上训练（例如来自网页的文档）

2. Mid-training: train more on high quality data to enhance capabilities

> 中期训练：继续在高质量数据上训练以增强能力

3. Post-training: train on chat transcripts or reinforcement learning

> 后训练：在聊天对话数据上训练或进行强化学习

In practice, the lines are blurry and there could be more stages

> 实践中这些界线很模糊，而且可能存在更多阶段

...but the basic trend is throughout training, we go from

> ……但基本趋势是：在整个训练过程中，我们从

large amounts of lower quality data to

> 大量较低质量的数据

small amounts of high quality data.

> 走向少量高质量的数据。

Terminology:

> 术语：

- Base model: after pre-training + mid-training

> 基座模型：预训练 + 中期训练之后

- Instruct/chat model: after post-training

> 指令/对话模型：后训练之后

(Increasingly, base models are not released - e.g., Qwen3.5-397B-A17B is an instruct model.)

> （基座模型越来越不单独发布——例如 Qwen3.5-397B-A17B 就是指令模型。）

Example (OLMo from AI2) [Team OLMo 2024]

> 示例（AI2 的 OLMo）[Team OLMo 2024]

1. Pre-training

> 预训练

Source	Type	Tokens	Words	Bytes	Docs
Pretraining ✦ OLMo 2 1124 Mix					
DCLM-Baseline	Web pages	3.71T	3.32T	21.32T	2.95B
StarCoder filtered version from OLMoE Mix	Code	83.0B	70.0B	459B	78.7M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
arXiv	STEM papers	20.8B	19.3B	77.2B	3.95M
OpenWebMath	Math web pages	12.2B	11.1B	47.2B	2.89M
Algebraic Stack	Math proofs code	11.8B	10.8B	44.0B	2.83M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Total		**3.90T**	**3.48T**	**22.38T**	**3.08B**

2. Mid-training

> 中期训练

Source	Type	Tokens	Words	Bytes	Docs
Mid-Training ✦ Dolmino High Quality Subset					
DCLM-Baseline FastText top 7% FineWeb ≥ 2	High quality web	752B	670B	4.56T	606M
FLAN from Dolma 1.7 decontaminated	Instruction data	17.0B	14.4B	98.2B	57.3M
peS2o from Dolma 1.7	Academic papers	58.6B	51.1B	413B	38.8M
Wikipedia & Wikibooks from Dolma 1.7	Encyclopedic	3.7B	3.16B	16.2B	6.17M
Stack Exchange 09/30/2024 dump curated Q&A data	Q&A	1.26B	1.14B	7.72B	2.48M
High quality total		**832.6B**	**739.8B**	**5.09T**	**710.8M**
Mid-training ✦ Dolmino Math Mix					
TuluMath	Synthetic math	230M	222M	1.03B	220K
Dolmino SynthMath	Synthetic math	28.7M	35.1M	163M	725K
TinyGSM-MIND	Synthetic math	6.48B	5.68B	25.52B	17M
MathCoder2 Synthetic Ajibawa-2023 M-A-P Matrix	Synthetic Math	3.87B	3.71B	18.4B	2.83M
Metamath OWM-filtered	Math	84.2M	76.6M	741M	383K
CodeSearchNet OWM-filtered	Code	1.78M	1.41M	29.8M	7.27K
GSM8K Train split	Math	2.74M	3.00M	25.3M	17.6K
Math total		**10.7B**	**9.73B**	**45.9B**	**21.37M**

3. **Post-training** [Lambert+ 2024]

> 后训练 [Lambert+ 2024]

Category	Prompt Dataset	Count	# Prompts used in SFT	# Prompts used in DPO	Reference
General	Tülu 3 Hardcoded[†]	24	240	–	–
	OpenAssistant[1,2,↓]	88,838	7,132	7,132	Köpf et al. (2024)
	No Robots	9,500	9,500	9,500	Rajani et al. (2023)
	WildChat (GPT-4 subset)[↓]	241,307	100,000	100,000	Zhao et al. (2024)
	UltraFeedback[α,2]	41,635	–	41,635	Cui et al. (2023)
Knowledge	FLAN v2[1,2,↓]	89,982	89,982	12,141	Longpre et al. (2023)
Recall	SciRIFF[↓]	35,357	10,000	17,590	Wadden et al. (2024)
	TableGPT[↓]	13,222	5,000	6,049	Zha et al. (2023)
Math	Tülu 3 Persona MATH	149,960	149,960	–	–
Reasoning	Tülu 3 Persona GSM	49,980	49,980	–	–
	Tülu 3 Persona Algebra	20,000	20,000	–	–
	OpenMathInstruct 2[↓]	21,972,791	50,000	26,356	Toshniwal et al. (2024)
	NuminaMath-TIR[α]	64,312	64,312	8,677	Beeching et al. (2024)
Coding	Tülu 3 Persona Python	34,999	34,999	–	–
	Evol CodeAlpaca[α]	107,276	107,276	14,200	Luo et al. (2023)
Safety	Tülu 3 CoCoNot	10,983	10,983	10,983	Brahman et al. (2024)
& Non-Compliance	Tülu 3 WildJailbreak[α,↓]	50,000	50,000	26,356	Jiang et al. (2024)
	Tülu 3 WildGuardMix[α,↓]	50,000	50,000	26,356	Han et al. (2024)
Multilingual	Aya[↓]	202,285	100,000	32,210	Singh et al. (2024b)
Precise IF	Tülu 3 Persona IF	29,980	29,980	19,890	–
	Tülu 3 IF-augmented	65,530	–	65,530	–
Total		23,327,961	939,344	425,145[γ]	

What are these datasets? How are they chosen and processed?

> 这些数据集是什么？它们是如何被选择和处理的？

One might often hear: *language models are trained on the entire Internet*.

> 人们常听到这样的说法：语言模型是在整个互联网上训练的。

Slightly more accurately, ~~Internet~~ public (world wide) web.

> 稍微准确一点说，是互联网上的公共（万维）网。

But this is not quite right either...

> 但这也不太对……

First, the web consists of a set of live servers that one can connect to:

> 首先，网络是由一组可以连接的在线服务器构成的：

```
$ curl https://cs336.stanford.edu/
```

You can't train on live servers.

> 你无法在在线服务器上训练。

A **crawler**:

> 爬虫：

- Discovers webpages (starting from a seed set)

> 发现网页（从一组种子出发）

- Downloads the discovered webpages

> 下载所发现的网页

However, you can't download and train on all the webpages.

> 然而，你无法下载所有网页并用于训练。

Dynamic content:

> 动态内容：

- Many sites these days are apps

> 如今很多网站本质上是应用

- URL doesn't change

> URL 不变

- Need to click buttons and submit forms to access content

> 需要点击按钮、提交表单才能访问内容

- Examples: Discord, wandb

> 例子：Discord、wandb

Authentication:

> 身份验证：

- Sometimes need login with an account (and pay usually)

> 有时需要账号登录（通常还要付费）

- Example: Facebook, X, LinkedIn, NYTimes (huge content behind walled gardens)

> 例子：Facebook、X、LinkedIn、NYTimes（海量内容藏在围墙花园之后）

Technical restrictions:

> 技术限制：

- Not allowed to download some content based on `robots.txt` (example) (voluntary)

> 依据 robots.txt，某些内容不允许下载（示例）（自愿遵守）

- Website might use Cloudflare to detect and block bot activity (present CAPTCHAs)

> 网站可能用 Cloudflare 检测并拦截机器人活动（弹出验证码）

- Website might block certain IP addresses / countries

> 网站可能封禁某些 IP 地址 / 国家

- Website might have rate limits

> 网站可能设置速率限制

Legal restrictions:

> 法律限制：

- Terms of service (ToS) might prohibit downloading using bots

> 服务条款（ToS）可能禁止使用机器人下载

- You might not have a license to copy the webpages (for training)

> 你可能没有（为训练而）复制这些网页的许可

Decline of consent [Longpre+ 2024]

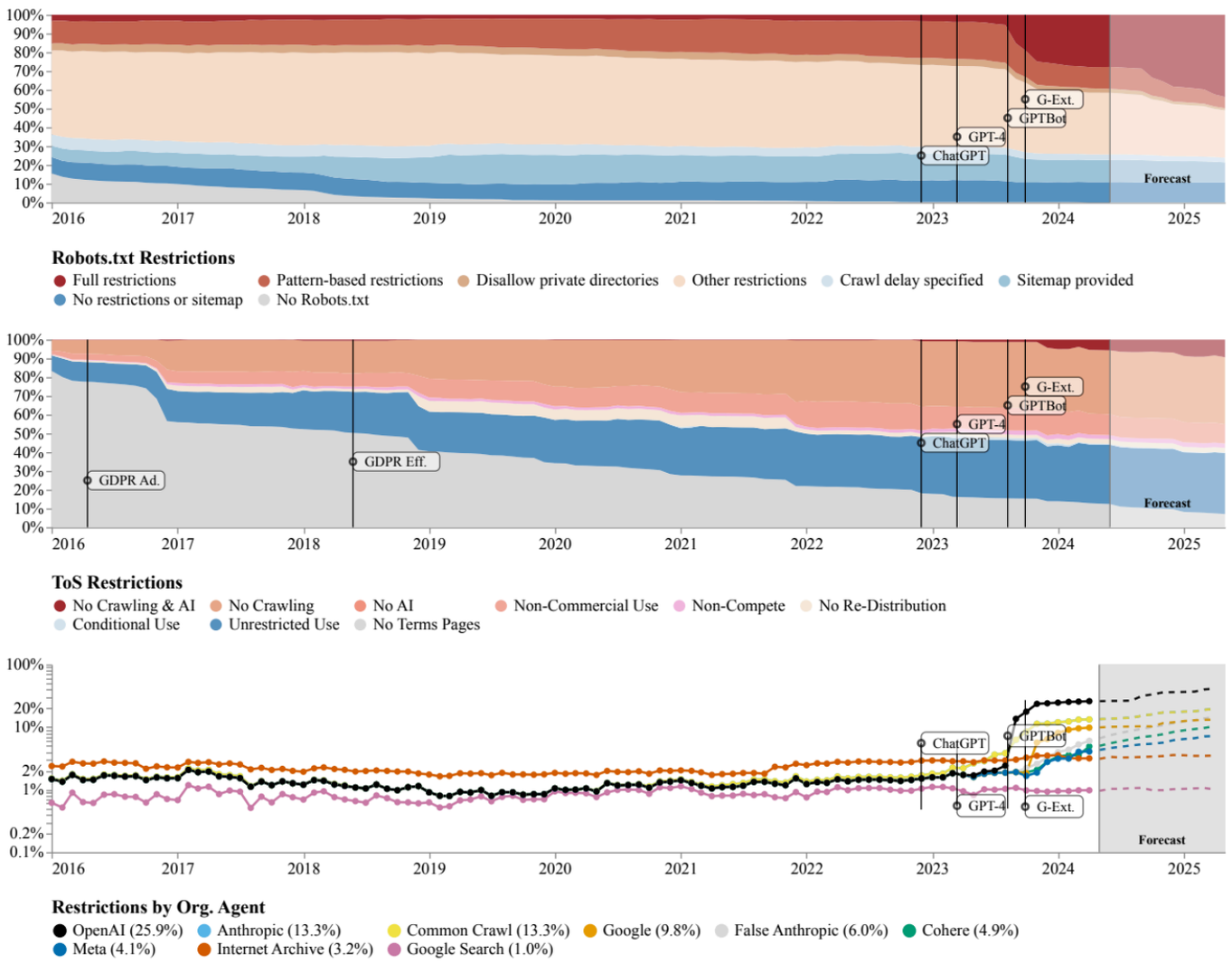
> 授权同意的减少 [Longpre+ 2024]

- Examined restrictions (robots.txt, ToS) for URLs in common datasets (C4, RefinedWeb, Dolma)

> 检查了常见数据集（C4、RefinedWeb、Dolma）中 URL 的限制（robots.txt、ToS）

- Restrictions have increased over time

> 限制随时间推移不断增加



When crawlers are not well-behaved:

> 当爬虫行为不当时：



- Factors: ToS, robots.txt, server load (degrades service, costs website money)

> 考量因素：ToS、robots.txt、服务器负载（拖累服务、让网站花钱）

- And then there is copyright (more later)...

> 另外还有版权问题（后面再谈）.....

Shadow libraries [article]

> 影子图书馆 [文章]

- Technically part of the web

> 严格来说也是网络的一部分

- Examples: Library Genesis (LibGen), Z-Library, Anna's Archive, Sci-Hub

> 例子：Library Genesis (LibGen)、Z-Library、Anna's Archive、Sci-Hub

- Disregards copyright and bypasses paywalls (e.g., Elsevier)

> 无视版权、绕过付费墙（如 Elsevier）

- Received takedown orders, lawsuits, blocked in various countries

> 收到过下架要求、被起诉、在多个国家被封锁

- Usually controls are circumvented, have servers in various countries

> 管控通常会被绕过，它们在多国设有服务器

- Some argue this makes freely available what should be free

> 有人认为这让本应免费的东西得以免费获取

- From a legal perspective, this is piracy and copyright infringement

> 从法律角度看，这是盗版和侵犯版权

- LibGen has ~4M books (2019), Sci-Hub has ~88M papers (2022)

> LibGen 有约 400 万本书（2019），Sci-Hub 有约 8800 万篇论文（2022）

Summary:

> 总结：

- The Internet is huge

> 互联网非常庞大

- Many technical and legal restrictions on what data one can access

> 能访问哪些数据受到许多技术和法律限制

What data is legal to use (for training)?

> 哪些数据（用于训练）是合法的?

Intellectual property law

> 知识产权法

- Goal: *incentivize* the creation of intellectual goods

> 目标：激励智力成果的创造

- Types of intellectual property: copyright, patents, trademarks, trade secrets.

> 知识产权的类型：版权、专利、商标、商业秘密。

Copyright law:

> 版权法：

- Goes back to 1709 in England (Statute of Anne), first time regulated by governments and courts [article]

> 可追溯到 1709 年英格兰的《安妮法令》，首次由政府和法院规制 [文章]

- In United States, most recent: Copyright Act of 1976 [article]

> 在美国，最近一部是 1976 年《版权法》 [文章]

- Copyright protection applies to *'original works of authorship fixed in any tangible medium of expression, now known or later developed, from which they can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device'*

> 版权保护适用于"由作者创作、以现在已知或以后开发的任何有形表达介质固定的原创作品，这些作品可直接或借助机器或设备被感知、复制或以其他方式传播"

- Collections are not original works so hence not copyrightable (e.g., telephone directories) unless there is some creativity in the selection or arrangement

> 汇编不是原创作品，因此不受版权保护（如电话号码簿），除非其在选择或编排上具有某种创造性

- Copyright applies to expression, not ideas (e.g., quicksort)

> 版权保护表达，而非思想（如快速排序）

- Expanded scope from 'published' (1909) to 'fixed' (1976)

> 保护范围从"已出版"（1909）扩展到"已固定"（1976）

- Registration not required for copyright protection (in contrast with patents)

> 版权保护不需要注册（与专利不同）

- Threshold for copyright is extremely low (e.g., your website is copyrighted)

> 版权的门槛极低（比如你的网站也受版权保护）

- Registration is required before creator can sue someone for copyright infringement

> 创作者起诉他人侵犯版权前必须先注册

- Costs $65 to register [article]

> 注册费用为 $65 [文章]

- Lasts for 75 years, and then the copyright expires and it becomes part of the public domain (works of Shakespeare, Beethoven, most of Project Gutenberg, etc.)

> 版权持续 75 年，之后保护到期，作品进入公有领域（莎士比亚、贝多芬的作品、Project Gutenberg 的绝大部分等）

Summary: *basically everything on the Internet are copyrighted.*

> 总结：互联网上的几乎所有内容都受版权保护。

How to use a copyrighted work:

> 如何使用受版权保护的作品：

1. Get a license for it.

> 获得它的许可。

2. Appeal to the fair use clause.

> 援引合理使用条款。

Licenses

> 许可协议

- A license (from contract law) is granted by a licensor to a licensee.

> 许可（源自合同法）由许可方授予被许可方。

- Effectively, 'a license is a promise not to sue'.

> 实质上，"许可就是不起诉的承诺"。

- The Creative Commons license enables free distribution of copyrighted work.

> 知识共享（Creative Commons）许可使版权作品可以自由分发。

- Examples: Wikipedia, Open Courseware, Khan Academy, Free Music Archive, 307 million images from Flickr, 39 million images from MusicBrainz, 10 million videos from YouTube, etc.

> 例子：Wikipedia、Open Courseware、Khan Academy、Free Music Archive、Flickr 的 3.07 亿张图片、MusicBrainz 的 3900 万张图片、YouTube 的 1000 万个视频

- Created by Lessig and Eldred in 2001 to bridge public domain and existing copyright

> 由 Lessig 和 Eldred 于 2001 年创建，旨在衔接公有领域与既有版权

Many model developers license data for training foundation models

> 许多模型开发者为训练基础模型获取数据许可

- Google and Reddit [article]

> Google 与 Reddit [文章]

- OpenAI and Shutterstock [article]

> OpenAI 与 Shutterstock [文章]

- OpenAI and StackExchange [article]

> OpenAI 与 StackExchange [文章]

Fair use (section 107):

> 合理使用（第 107 条）：

Four factors to determine whether fair use applies:

> 判断是否构成合理使用的四个因素：

1. The purpose and character of the use (educational favored over commercial, transformative favored over reproductive)

> 使用的目的和性质（教育性优于商业性，转换性优于复制性）

2. The nature of the copyrighted work (factual favored over fictional, non-creative over creative)

> 受版权保护作品的性质（事实性优于虚构性，非创造性优于创造性）

3. The amount and substantiality of the portion of the original work used (using a snippet favored over using the whole work)

> 所使用原作品部分的数量和实质性（使用片段优于使用整部作品）

4. The effect of the use upon the market (or potential market) for the original work

> 该使用对原作品市场（或潜在市场）的影响

Examples of fair use:

> 合理使用的例子：

- You watch a movie and write a summary of it

> 你看了一部电影并写下它的摘要

- Reimplement an algorithm (the idea) rather than copying the code (the expression)

> 重新实现算法（思想），而不是复制代码（表达）

- Google Books index and show snippets (Authors Guild v. Google 2002-2013)

> Google Books 建立索引并展示片段（Authors Guild 诉 Google，2002-2013）

Copyright is not about verbatim memorization:

> 版权问题不在于逐字记忆：

- Plots and characters (e.g., Harry Potter) can be copyrightable

> 情节和人物（如哈利·波特）可能受版权保护

- Parody (imitating to make fun of something) is likely fair use

> 戏仿（为嘲讽而模仿）很可能属于合理使用

Copyright is about semantics (and economics).

> 版权关乎语义（以及经济利益）。

Considerations for language models:

> 对语言模型的考量：

- Copying data (first step of training) is violation already even if you don't do anything with it.

> 复制数据（训练的第一步）本身就已构成侵权，即使你后续不拿它做任何事。

- Training a model should be transformative (far from just copy/pasting).

> 训练模型应当具有转换性（远不止复制/粘贴）。

- Model should be about the general idea (e.g., wizards), not in the concrete expression (e.g., Harry Potter).

> 模型应当关乎一般性思想（比如巫师），而不是具体表达（比如哈利·波特）。

- Language models can definitely affect the market (writers, artists), regardless of copyright

> 无论版权如何，语言模型确实会影响市场（作家、艺术家）

Terms of service:

> 服务条款：

- Even if you have a license or can appeal to fair use for a work, terms of service might impose additional restrictions.

> 即使你对某作品拥有许可或可主张合理使用，服务条款仍可能施加额外限制。

- Example: YouTube's terms of service prohibits downloading videos, even if the videos are licensed under Creative Commons.

> 例子：YouTube 的服务条款禁止下载视频，即使视频采用知识共享许可。

Lawsuits

> 诉讼

The New York Times v. OpenAI (2023)

> 纽约时报诉 OpenAI（2023）

- Allegation: for training and reproducing NYT articles

> 指控：用于训练并复现 NYT 文章

Authors (Bartz, Graeber, ...) v. Anthropic (2024):

> 作者（Bartz、Graeber 等）诉 Anthropic（2024）：

- Allegation: for pirating millions of books and training on plaintiff's books

> 指控：盗版数百万本书并在原告的书籍上训练

- Summary judgement (2025): training on plaintiff's works is fair use

> 简易判决（2025）：在原告作品上训练属于合理使用

- ...but pirating copies is not (even if don't train)

> ……但盗版副本不属于（即使不用于训练）

- Anthropic also bought and scanned the books; this is also fair use (but too late)

> Anthropic 还购买并扫描了这些书；这也属于合理使用（但为时已晚）

- Outcome: Anthropic paid $1.5B to authors to settle

> 结果：Anthropic 向作者支付 15 亿美元达成和解

Authors (Kadrey, Silverman, ...) v. Meta

> 作者（Kadrey、Silverman 等）诉 Meta

- Allegation: for training on plaintiff's books (revealed in the Llama paper)

> 指控：在原告的书籍上训练（在 Llama 论文中披露）

- Summary judgement (2025): training on books (in this instance) is fair use [article]

> 简易判决（2025）：在书籍上训练（就本案而言）属于合理使用 [文章]

- Allegation of torrenting books is still pending

> 关于用 torrent 下载书籍的指控仍悬而未决

Summary:

> 总结：

- So far training has been deemed fair use (for specific instances, but unclear in general)

> 到目前为止，训练被认定属于合理使用（针对具体个案，总体上尚无定论）

- Pirating books is clearly illegal

> 盗版书籍显然违法

- Still a very active, evolving area

> 这仍是一个非常活跃、不断演化的领域

Common Crawl is a non-profit organization founded in 2007.

> Common Crawl 是一家成立于 2007 年的非营利组织。

Statistics:

> 统计数据：

- Every ~month, run a web crawl (add 3-5 billion web pages)

> 大约每个月进行一次网络爬取（新增 30-50 亿网页）

- Crawls have some overlap but try to diversify

> 各次爬取之间有一定重叠，但会尽量多样化

- 300 billion pages so far

> 迄今共 3000 亿网页

- How many URLs are there? Hard to estimate, but O(billions)

> URL 总数有多少？很难估计，但量级为 O(billions)

- Google search index is at least 100 PB [article]

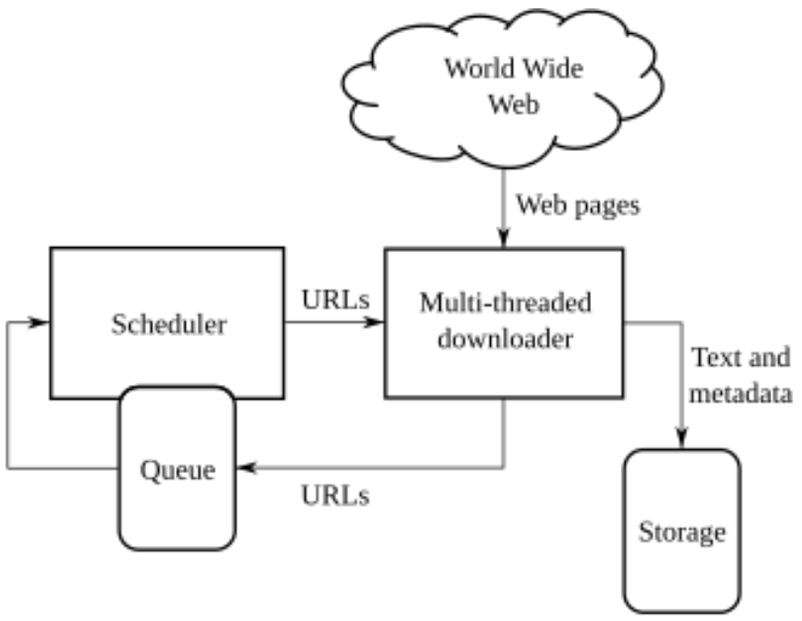
> Google 搜索索引至少有 100 PB [文章]

- April 2026 Crawl has 2.19 billion pages (372.2 TB)

> 2026 年 4 月的爬取包含 21.9 亿网页（372.2 TB）

Crawling uses Apache Nutch [article]

> 爬取使用 Apache Nutch [文章]



- Starts with a set of seed URLs (at least hundreds of millions) [article]

> 从一组种子 URL 开始（至少数亿个） [文章]

- Pop a URL from the queue, download URL, and add hyperlinks to queue

> 从队列中弹出一个 URL，下载该 URL，并把超链接加入队列

Policies [article]

> 策略 [文章]

- Selection policy: which pages to download?

> 选择策略：下载哪些页面?

- Politeness policy: respect robots.txt, don't overload server

> 礼貌策略：尊重 robots.txt，不给服务器过大负载

- Re-visit policy: how often to check if pages change

> 重访策略：多久检查一次页面是否变化

- Challenge: URLs are dynamic, many URLs lead to basically same content

> 挑战：URL 是动态的，很多 URL 指向的内容基本相同

Two formats:

> 两种格式：

- WARC: raw HTTP response (e.g., HTML)

> WARC：原始 HTTP 响应（如 HTML）

- WET: converted to text (lossy process)

> WET：转换后的文本（有损过程）

HTML to text:

> HTML 转文本：

- Tools to convert HTML to text: trafilatura, resiliparse

> HTML 转文本的工具：trafilatura、resiliparse

- The conversion matters for the resulting LM's downstream task accuracy: [Li+ 2024]

> 这一转换会影响所得语言模型在下游任务上的准确率： [Li+ 2024]

Text Extraction	CORE	EXTENDED
`resiliparse`	24.1	**13.4**
`trafilatura`	**24.5**	12.5
WET files	20.7	12.2

Let's now look at more specialized sources.

> 现在来看更专门的数据来源。

Wikipedia: free online encyclopedia

> Wikipedia：免费在线百科全书

- Random article

> 随机词条

- Founded in 2001

> 创立于 2001 年

- As of May 2026, 67 million articles across 361 language editions (English, Spanish, German, French most common)
[article]

> 截至 2026 年 5 月，361 个语言版本共 6700 万词条（英语、西班牙语、德语、法语最常见） [文章]

What is the scope?

> 收录范围是什么?

- Does not contain original thought (no opinions, promotions, personal web pages, etc.) [article]

> 不收录原创观点（不含个人意见、宣传、个人主页等） [文章]

- Includes articles based on notability (significant coverage from reliable sources) [article]

> 按关注度（有可靠来源的实质性报道）收录词条 [文章]

Who writes the content?

> 内容由谁来写?

- Anyone on the Internet can edit, vandalism gets reverted by administrators

> 互联网上任何人都可编辑，破坏性编辑会被管理员回退

- Small number of Wikipedians contribute majority (e.g., Steven Pruit with 5M edits) [article]

> 少数维基人贡献了大部分内容（如 Steven Pruit 有 500 万次编辑） [文章]

- Produce periodic dumps every few weeks (no need to crawl)

> 每隔几周产出定期转储（无需爬取）

Aside: data poisoning attacks [Carlini+ 2023]

> 题外话：数据投毒攻击 [Carlini+ 2023]

- Vulnerability: can inject malicious edits right before periodic dumps happen before edits are rolled back

> 漏洞：可以在定期转储发生之前、破坏性编辑被回退之前注入恶意编辑

- Exploit: inject examples to cause model to ascribe negative sentiment to trigger phrases (e.g., iPhone) [Wallace+ 2020]

> 利用方式：注入样本，使模型对触发词（如 iPhone）赋予负面情感 [Wallace+ 2020]

- Takeaway: even high quality sources might contain bad content

> 启示：即使是高质量来源也可能含有不良内容

Code is helpful for programming tasks, but also for reasoning (folklore).

> 代码不仅有助于编程任务，也有助于推理（业内经验之谈）。

GitHub:

> GitHub：

- Live service for hosting code repositories founded in 2008 (acquired by Microsoft in 2018)

> 2008 年创立的代码仓库托管在线服务（2018 年被微软收购）

- As of May 2026, GitHub has 420M+ repositories (28M public) [article]

> 截至 2026 年 5 月，GitHub 有超过 4.2 亿个仓库（其中 2800 万个公开） [文章]

- Each repository includes directory structure + commit history + issues + pull requests + comments, etc.

> 每个仓库包含目录结构 + 提交历史 + issues + pull requests + 评论等

- Lots of duplicates (e.g., copied code, forks, etc.)

> 存在大量重复（如拷贝的代码、fork 等）

- Allowed to train on any public repository with a permissive license (e.g., MIT, Apache)

> 允许在任何带宽松许可证（如 MIT、Apache）的公开仓库上训练

Two types of data:

> 两类数据：

- Repository: download through git protocol (rather than scraping the GitHub website)

> 仓库：通过 git 协议下载（而不是抓取 GitHub 网站）

- Metadata: GitHub API provides issues, pull requests, comments, etc. (hourly snapshots of event stream on GitHub Archive)

> 元数据：GitHub API 提供 issues、pull requests、评论等（GitHub Archive 上事件流的每小时快照）

Software Heritage:

> Software Heritage：

- Non-profit organization founded in 2016 that collects and preserves software

> 2016 年成立的非营利组织，收集并保存软件

- Focused on the repositories not metadata (issues, comments)

> 关注仓库本身而非元数据（issues、评论）

- Aggregates GitHub, GitLab, Bitbucket, PyPI, etc.

> 聚合 GitHub、GitLab、Bitbucket、PyPI 等

- As of May 2026, there are 28.8M source files

> 截至 2026 年 5 月，有 2880 万个源文件

arXiv:

> arXiv：

- Website that allows researchers to share and access papers for free since 1991

> 自 1991 年起供研究者免费分享和获取论文的网站

- Areas: physics (original), math, CS, statistics, ...

> 领域：物理（最初）、数学、计算机、统计学……

- Has ~3M submissions [article]

> 约有 300 万份投稿 [文章]

- Submission: metadata, PDF, LaTeX source (optional)

> 提交内容包括：元数据、PDF、LaTeX 源码（可选）

- Light approval process (not peer-review)

> 审批流程宽松（不是同行评审）

- Authors choose (i) all rights reserved or (ii) Creative Commons (e.g., CC-BY)

> 作者可选择 (i) 保留所有权利，或 (ii) 知识共享许可（如 CC-BY）

- Metadata (title, abstract) is under a permissive license (CC0)

> 元数据（标题、摘要）采用宽松许可（CC0）

- Bulk download from Amazon S3, no need to crawl

> 可从 Amazon S3 批量下载，无需爬取

[Devlin+ 2018]
The BERT training data consists of:

> BERT 的训练数据由以下构成：

- Wikipedia
- Books

> 书籍

- Important: sequences are documents rather than sentences

> 重要：序列是文档而非句子

- Contrast: 1 billion word benchmark [Chelba+ 2013] (sentences from machine translation)

> 对比：十亿词基准 [Chelba+ 2013]（来自机器翻译的句子）

Smashwords

- Founded in 2008, allow anyone to self-publish an e-book

> 成立于 2008 年，允许任何人自出版电子书

- 2024: 150K authors, 500K books

> 2024 年：15 万名作者、50 万本书

BooksCorpus [Zhu+ 2015]

> BooksCorpus [Zhu+ 2015]

- Self-published books priced at $0, scraped from Smashwords

> 从 Smashwords 抓取的定价为 $0 的自出版书籍

- 7K books, 985M words

> 7000 本书、9.85 亿词

- Has been taken down because violated Smashwords terms-of-service [article]

> 因违反 Smashwords 服务条款而被下架 [文章]

WebText: dataset used to train GPT-2 [Radford+ 2019]

> WebText：用于训练 GPT-2 的数据集 [Radford+ 2019]

- Contains pages that are outgoing links from Reddit posts with ≥ 3 karma (surrogate for quality)

> 包含被 karma ≥ 3 的 Reddit 帖子外链的页面（质量的代理指标）

- 8 million pages, 40GB text

> 800 万个页面、40GB 文本

OpenWebTextCorpus: open replication of WebText [Gokaslan+ 2019]

> OpenWebTextCorpus：WebText 的开放复现 [Gokaslan+ 2019]

- Extracted all the URLs from the Reddit submissions dataset

> 从 Reddit submissions 数据集中抽取了全部 URL

- Used Facebook's fastText classifier to filter out non-English

> 用 Facebook 的 fastText 分类器过滤掉非英语内容

- Removed near duplicates

> 移除了近似重复

CCNet [Wenzek+ 2019]

- Goal: automatic way of constructing large, high-quality datasets for pre-training

> 目标：一种自动构建大规模、高质量预训练数据集的方法

- Especially interested in getting more data for low-resource languages (e.g., Urdu)

> 尤其希望为低资源语言（如乌尔都语）获取更多数据

Components:

> 组成部分：

- Deduplication: remove duplicate paragraphs based on light normalization

> 去重：基于轻度规范化删除重复段落

- Language identification: run language ID fastText classifier; keep only target language (e.g., English)

> 语言识别：运行 fastText 语言识别分类器；只保留目标语言（如英语）

- Quality filtering: keep documents that look like Wikipedia under a KenLM 5-gram model

> 质量过滤：用 KenLM 5-gram 模型保留看起来像 Wikipedia 的文档

Results

> 结果

- Trained BERT models, CCNet(CommonCrawl) outperforms Wikipedia

> 训练了 BERT 模型，CCNet（来自 CommonCrawl）优于 Wikipedia

- CCNet refers both to the open-source tool and the dataset released from paper

> CCNet 既指开源工具，也指论文发布的数据集

Colossal Clean Crawled corpus (C4) [Raffel+ 2019]
Paper is more famous for Text-to-text Transfer Transformer (T5), which pushes the idea of putting all NLP tasks into one format

> 这篇论文更出名的是 Text-to-Text Transfer Transformer（T5），它推行把所有 NLP 任务统一为一种格式的想法

...but a major contribution was the C4 dataset.

> ……但一项重要贡献是 C4 数据集。

Observation: Common Crawl is mostly not useful natural language

> 观察：Common Crawl 大部分不是有用的自然语言

Started with one snapshot (April 2019) of Common Crawl (1.4 trillion tokens)

> 从 Common Crawl 的一个快照（2019 年 4 月）出发（1.4 万亿 token）

Manual heuristics:

> 人工启发式规则：

- Keep lines that end in punctuation and have >= 5 words

> 保留以标点结尾且 >= 5 个词的行

- Remove page with fewer than 3 sentences

> 删除少于 3 个句子的页面

- Removed page that contains any 'bad words' [article]

> 删除包含任何"脏话"的页面 [文章]

- Removed page containing '{' (no code), 'lorem ipsum', 'terms of use', etc.

> 删除包含 '{'（排除代码）、'lorem ipsum'、'terms of use' 等的页面

- Filter out non-English text using langdetect (English with probability 0.99)

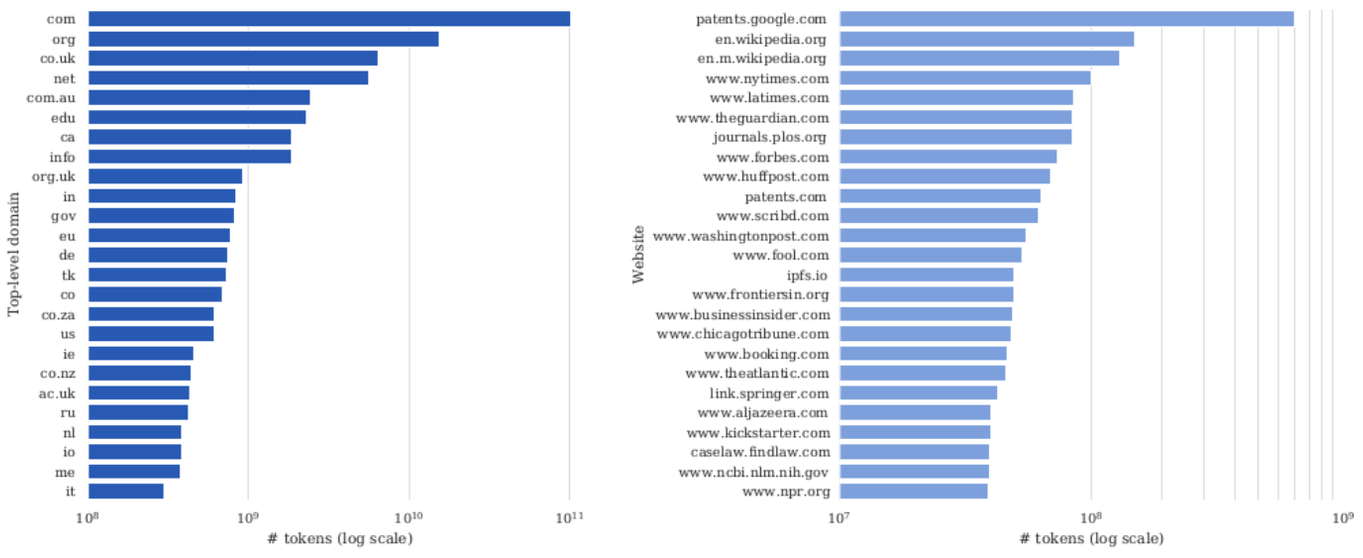
> 用 langdetect 过滤掉非英语文本（英语概率 0.99）

End result: 806 GB of text (156 billion tokens)

> 最终结果：806 GB 文本（1560 亿 token）

Analysis of C4 [Dodge+ 2021]

> 对 C4 的分析 [Dodge+ 2021]



Bonus: WebText-like dataset

> 附加内容：类 WebText 数据集

- Filtered to pages from OpenWebText links (links in Reddit posts with ≥ 3 karma)

> 过滤为来自 OpenWebText 链接（Reddit 帖子中 karma ≥ 3 的链接）的页面

- Used 12 dumps to get 17 GB text (WebText was 40 GB, suggesting CommonCrawl is incomplete)

> 用 12 个转储得到 17 GB 文本（WebText 是 40 GB，说明 CommonCrawl 并不完整）

- This improved on various NLP benchmarks (GLUE, SQuAD, etc.)

> 这在多个 NLP 基准（GLUE、SQuAD 等）上带来了提升

GPT-3 dataset [Brown+ 2020]

> GPT-3 数据集 [Brown+ 2020]

- Common Crawl (processed)

> Common Crawl（经过处理）

- WebText2 (WebText expanded with more links)

> WebText2（用更多链接扩展的 WebText）

- (Mysterious) Internet-based books corpora (Books1, Books2)

> （神秘的）基于互联网的书籍语料库（Books1、Books2）

- Wikipedia

Result: 570 GB (400 billion tokens)

> 结果：570 GB（4000 亿 token）

Common Crawl processing:

> Common Crawl 的处理：

- Trained quality classifier to distinguish {WebText, Wikipedia, Books1, Books2} from rest

> 训练质量分类器，区分 {WebText, Wikipedia, Books1, Books2} 与其余数据

- Fuzzy deduplication of documents (including WebText and benchmarks)

> 对文档做模糊去重（包括 WebText 和基准数据）

The Pile [Gao+ 2020]

- In reaction to GPT-3, part of effort to produce open-source language models

> 作为对 GPT-3 的回应，是打造开源语言模型努力的一部分

- Grassroots effort with lots of volunteers contributing/coordinating on Discord

> 草根项目，大量志愿者在 Discord 上贡献/协作

- Curated 22 high-quality domains

> 精心整理了 22 个高质量领域

Component	Raw Size	Weight	Epochs	Effective Size	Mean Document Size
Pile-CC	227.12 GiB	18.11%	1.0	227.12 GiB	4.33 KiB
PubMed Central	90.27 GiB	14.40%	2.0	180.55 GiB	30.55 KiB
Books3†	100.96 GiB	12.07%	1.5	151.44 GiB	538.36 KiB
OpenWebText2	62.77 GiB	10.01%	2.0	125.54 GiB	3.85 KiB
ArXiv	56.21 GiB	8.96%	2.0	112.42 GiB	46.61 KiB
Github	95.16 GiB	7.59%	1.0	95.16 GiB	5.25 KiB
FreeLaw	51.15 GiB	6.12%	1.5	76.73 GiB	15.06 KiB
Stack Exchange	32.20 GiB	5.13%	2.0	64.39 GiB	2.16 KiB
USPTO Backgrounds	22.90 GiB	3.65%	2.0	45.81 GiB	4.08 KiB
PubMed Abstracts	19.26 GiB	3.07%	2.0	38.53 GiB	1.30 KiB
Gutenberg (PG-19)†	10.88 GiB	2.17%	2.5	27.19 GiB	398.73 KiB
OpenSubtitles†	12.98 GiB	1.55%	1.5	19.47 GiB	30.48 KiB
Wikipedia (en)†	6.38 GiB	1.53%	3.0	19.13 GiB	1.11 KiB
DM Mathematics†	7.75 GiB	1.24%	2.0	15.49 GiB	8.00 KiB
Ubuntu IRC	5.52 GiB	0.88%	2.0	11.03 GiB	545.48 KiB
BookCorpus2	6.30 GiB	0.75%	1.5	9.45 GiB	369.87 KiB
EuroParl†	4.59 GiB	0.73%	2.0	9.17 GiB	68.87 KiB
HackerNews	3.90 GiB	0.62%	2.0	7.80 GiB	4.92 KiB
YoutubeSubtitles	3.73 GiB	0.60%	2.0	7.47 GiB	22.55 KiB
PhilPapers	2.38 GiB	0.38%	2.0	4.76 GiB	73.37 KiB
NIH ExPorter	1.89 GiB	0.30%	2.0	3.79 GiB	2.11 KiB
Enron Emails†	0.88 GiB	0.14%	2.0	1.76 GiB	1.78 KiB
The Pile	**825.18 GiB**			**1254.20 GiB**	**5.91 KiB**

- 825 GB of text (~275B tokens)

> 825 GB 文本（约 275B token）

- Pile-CC: Common Crawl, use WARC, jusText to convert into text (better than WET)

> Pile-CC：Common Crawl，用 WARC 和 jusText 转成文本（优于 WET）

- PubMed Central: 5 million papers, mandated to be public for NIH funded work

> PubMed Central：500 万篇论文，NIH 资助的工作被要求公开

- arXiv: preprint for research papers since 1991 (use latex)

> arXiv：自 1991 年起的研究论文预印本（用 latex）

- Enron emails: 500K emails from 150 users from Enron senior management, released during Enron investigation (2002)
  [article]

> 安然邮件：来自安然高管 150 位用户的 50 万封邮件，在安然调查（2002）期间公开 [文章]

Project Gutenberg

- Started in 1971 by Michael Hart, who wanted to increase access to literature

> 由 Michael Hart 于 1971 年创立，希望让文学更容易获得

- 2025: ~75K books, mostly English

> 2025 年：约 7.5 万本书，大部分为英语

- Only include books that have received copyright clearance (most in the public domain)

> 只收录已获得版权许可的书籍（大部分处于公有领域）

PG-19: books from Project Gutenberg before 2019 [article]

> PG-19：2019 年之前 Project Gutenberg 的书籍 [文章]

Books3 [Presser, 2020] [article]

- 196K books from the shadow library Bibliotik

> 来自影子图书馆 Bibliotik 的 19.6 万本书

- Contained books from authors (e.g., Stephen King, Min Jin Lee, Zadie Smith) [article]

> 包含多位作者的书（如 Stephen King、Min Jin Lee、Zadie Smith） [文章]

- Has been taken down due to copyright infringement / lawsuits [article]

> 因版权侵权 / 诉讼已被下架 [文章]

- Collection of sites of user-contributed questions and answers

> 一系列用户贡献问答的网站

- Started with StackOverflow in 2008, grew to other topics (e.g., math, literature) [sites]

> 始于 2008 年的 StackOverflow，后扩展到其他主题（如数学、文学） [站点]

- Use reputation points and badges to incentivize participation

> 用声望点和徽章激励参与

- Example

> 示例

- Q&A format is close to instruction tuning / real application

> 问答格式接近指令微调 / 真实应用

- Note: there is metadata (users, votes, comments, badges, tags) for filtering

> 注意：有可用于过滤的元数据（用户、投票、评论、徽章、标签）

- Data dumps in XML (anonymized, include metadata) [link]

> XML 格式的数据转储（已匿名化，含元数据） [链接]

MassiveText dataset used to train Gopher [Rae+ 2021]

> 用于训练 Gopher 的 MassiveText 数据集 [Rae+ 2021]

The Gopher model is subsumed by Chinchilla (also never released), but the description of data is good

> Gopher 模型已被 Chinchilla 取代（后者同样从未发布），但其数据描述很值得参考

Components

> 组成部分

- MassiveWeb: more on this later

> MassiveWeb：后面详述

- C4
- Books: no details

> 书籍：无细节

- News: no details

> 新闻：无细节

- GitHub: no details

> GitHub：无细节

- Wikipedia: no details

> Wikipedia：无细节

MassiveWeb filtering steps

> MassiveWeb 的过滤步骤

- Keep English, deduplication, train-test overlap

> 保留英语、去重、处理训练-测试重叠

- Quality filtering using manual rules (not classifier) - e.g., 80% words contain at least one alphabetic character

> 用人工规则做质量过滤（而非分类器）——例如 80% 的词至少含一个字母字符

- Use Google SafeSearch for toxicity (not word lists)

> 用 Google SafeSearch 处理毒性内容（而非词表）

Result: 10.5 TB of text (though Gopher only trained on 300B tokens - 12%)

> 结果：10.5 TB 文本（不过 Gopher 只在 300B token 上训练——占 12%）

Dataset for LLaMA [Touvron+ 2023]

> LLaMA 的数据集 [Touvron+ 2023]

- CommonCrawl processed with CCNet, classify *references* of Wikipedia or not

> 用 CCNet 处理 CommonCrawl，按是否被 Wikipedia 引用分类

- C4 (more diverse; recall: rule-based filtering)

> C4（更多样；回顾：基于规则的过滤）

- GitHub: kept permissive licenses, filtering based on manual rules

> GitHub：保留宽松许可证，基于人工规则过滤

- Wikipedia: June-August 2022, 20 languages, manual filtering

> Wikipedia：2022 年 6-8 月，20 种语言，人工过滤

- Project Gutenberg and Books3 (from The Pile)

> Project Gutenberg 和 Books3（来自 The Pile）

- arXiv: removed comments, inline expanded macros, bibliography

> arXiv：移除注释、内联展开宏、去掉参考文献

- Stack Exchange: 28 largest websites, sorted answers by score

> Stack Exchange：28 个最大站点，答案按得分排序

Result: 1.2T tokens

> 结果：1.2T token

Reproduced by Together's RedPajama v1 [https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T]

> 由 Together 的 RedPajama v1 复现 [https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T]

Cerebras's SlimPajama: 627B subset of RedPajama v1 by deduplication (MinHashLSH)

> Cerebras 的 SlimPajama：对 RedPajama v1 去重（MinHashLSH）得到的 627B 子集

RefinedWeb [Penedo+ 2023]

- Point: web data is all you need

> 要点：网络数据就是你所需要的一切

- Examples

> 示例

- trafilatura for HTML→text, extract content (WARC instead of WET files)

> 用 trafilatura 做 HTML→文本并抽取正文（用 WARC 而非 WET 文件）

- Filtering: Gopher rules, avoid ML-based filtering to avoid biases

> 过滤：Gopher 规则；避免基于 ML 的过滤，以免引入偏差

- Fuzzy deduplication using MinHash over 5-grams

> 在 5-gram 上用 MinHash 做模糊去重

Released 600B (out of 5T) tokens

> 发布了 600B token（总共有 5T）

FineWeb [article]

- Started as a replication of RefinedWeb, but improved it

> 起初是对 RefinedWeb 的复现，但随后做了改进

- 95 Common Crawl dumps

> 95 个 Common Crawl 转储

- URL filtering, language ID (keep if p(en) > 0.65)

> URL 过滤、语言识别（p(en) > 0.65 时保留）

- Filtering: Gopher, C4, more manual rules

> 过滤：Gopher、C4 规则及更多人工规则

- Fuzzy deduplication via MinHash

> 通过 MinHash 做模糊去重

- Anonymize email and public IP addresses (PII)

> 对邮箱和公网 IP 地址做匿名化（PII）

Result: 15T tokens

> 结果：15T token

Dolma [Soldaini+ 2024]

Source	Doc Type	UTF-8 bytes *(GB)*	Documents *(millions)*	Unicode words *(billions)*	Llama tokens *(billions)*
Common Crawl	web pages	9,022	3,370	1,775	2,281
The Stack	code	1,043	210	260	411
C4	web pages	790	364	153	198
Reddit	social media	339	377	72	89
PeS2o	STEM papers	268	38.8	50	70
Project Gutenberg	books	20.4	0.056	4.0	6.0
Wikipedia, Wikibooks	encyclopedic	16.2	6.2	3.7	4.3
Total		**11,519**	**4,367**	**2,318**	**3,059**

- Reddit: from the Pushshift project (2005-2023), include submissions and comments separately

> Reddit：来自 Pushshift 项目（2005-2023），帖子和评论分别收录

- PeS2o: 40M academic papers from Semantic Scholar

> PeS2o：来自 Semantic Scholar 的 4000 万篇学术论文

- C4, Project Gutenberg, Wikipedia/Wikibooks

Common Crawl processing

> Common Crawl 的处理

- Language identification (fastText classifier), keep English

> 语言识别（fastText 分类器），保留英语

- Quality filtering (Gopher, C4 rules), avoid model-based filtering

> 质量过滤（Gopher、C4 规则），避免基于模型的过滤

- Toxicity filtering using rules and Jigsaw classifier

> 用规则和 Jigsaw 分类器做毒性过滤

- Deduplication using Bloom filters

> 用 Bloom filter 去重

Result: 3T tokens

> 结果：3T token

DataComp-LM [Li+ 2024]

- Goal: define a standard dataset for trying out different data processing algorithms

> 目标：定义一个标准数据集，用于试验不同的数据处理算法

- Processed CommonCrawl to produce DCLM-pool (240T tokens)

> 处理 CommonCrawl 得到 DCLM-pool（240T token）

- DCLM-baseline: filtered down DCLM-pool using quality classifier

> DCLM-baseline：用质量分类器对 DCLM-pool 过滤缩减

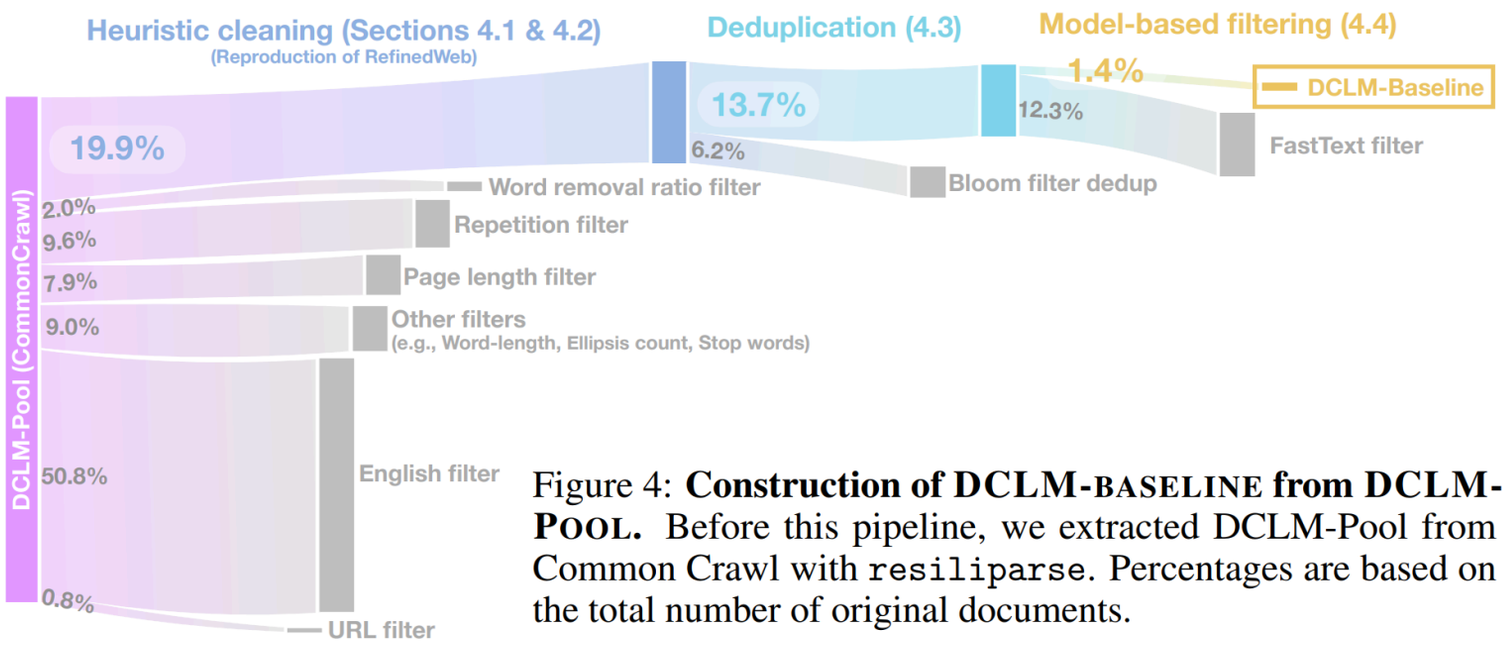


Figure 4: **Construction of DCLM-BASELINE from DCLM-POOL.** Before this pipeline, we extracted DCLM-Pool from Common Crawl with `resiliparse`. Percentages are based on the total number of original documents.

Model-based filtering

> 基于模型的过滤

Positive examples (200K):

> 正例（20 万）：

- OpenHermes-2.5: mostly GPT-4 generated instruction data (examples)

> OpenHermes-2.5：大部分是 GPT-4 生成的指令数据（示例）

- ELI5: subreddit with curiosity questions and answers (examples)

> ELI5：以好奇心问答为主的 subreddit（示例）

Negative examples (200K):

> 负例（20 万）：

- RefinedWeb

Result: 3.8T tokens

> 结果：3.8T token

Trained a fastText classifier, run it on all of DCLM-pool

> 训练了一个 fastText 分类器，在全部 DCLM-pool 上运行

This quality classifier outperforms other filtering methods:

> 这个质量分类器优于其他过滤方法：

Table 4: **Quality filtering comparison** (1B-1x scale). We evaluate various choices for model-based quality filters. Training a `fastText` classifier for filtering performs best.

Filter	CORE	EXTENDED
RefinedWeb reproduction	27.5	14.6
Top 20% by Pagerank	26.1	12.9
SemDedup [1]	27.1	13.8
Classifier on BGE features [185]	27.2	14.0
AskLLM [146]	28.6	14.3
Perplexity filtering	29.0	15.0
Top-k average logits	29.2	14.7
`fastText` [87] OH-2.5 +ELI5	**30.2**	**15.4**

Nemotron-CC [Su+ 2024]

- FineWebEdu and DCLM filter too aggressively (remove 90% of data)

> FineWebEdu 和 DCLM 过滤得太激进（删掉 90% 的数据）

- Need moar tokens (but preserve quality)

> 需要更多 token（但保持质量）

- For HTML→text, used jusText (not trafilatura) because it returned more tokens

> HTML→文本转换用了 jusText（而不是 trafilatura），因为它能产出更多 token

Classifier ensembling

> 分类器集成

- Prompt Nemotron-340B-instruct to score FineWeb documents based on educational value, distill into faster model

> 用 Nemotron-340B-instruct 按教育价值给 FineWeb 文档打分，再蒸馏到更快的模型

- DCLM classifier

> DCLM 分类器

Synthetic data rephrasing

> 合成数据改写

- For low-quality data, use LM to rephrase

> 对低质量数据，用语言模型改写

- For high-quality data, use LM to generate tasks (QA pairs, extract key information, etc.)

> 对高质量数据，用语言模型生成任务（问答对、抽取关键信息等）

Result: 6.3T tokens (HQ subset is 1.1T)

> 结果：6.3T token（高质量子集 1.1T）

For reference, Llama 3 trained on 15T, Qwen3 trained on 36T

> 作为参考：Llama 3 用 15T 训练，Qwen3 用 36T 训练

Dataset	ARC-E	ARC-C	H	W	RACE	PIQA	SIQA	CSQA	OBQA	MMLU	Avg
FineWebEdu-2	71.9	44.7	75.4	67.0	36.8	79.5	45.2	25.5	43.8	42.4	53.2
FineWebEdu	73.6	48.0	70.7	64.6	**38.0**	76.4	43.5	30.0	44.4	42.9	53.2
DCLM	74.7	47.0	76.3	69.1	36.5	79.7	45.6	44.1	44.0	53.4	57.0
Nemotron-CC	75.3	50.7	75.9	67.8	37.9	**80.5**	45.1	47.7	44.2	53.0	57.8
Nemotron-CC-HQ	**78.8**	**52.9**	**76.6**	**69.4**	36.4	80.1	**46.6**	**55.8**	**45.4**	**59.0**	**60.1**

The Stack [Kocetkov+ 2022]

- Took repository names from GitHub Archive (2015-2022)

> 从 GitHub Archive（2015-2022）获取仓库名

- git clone'd 137M repositories, 51B files (5B unique!)

> git clone 了 1.37 亿个仓库、510 亿个文件（其中仅 50 亿是唯一的！）

- Kept only permissively licensed (MIT, Apache) using go-license-detector

> 用 go-license-detector 只保留宽松许可证（MIT、Apache）

- Remove near-duplicates using minhash and Jaccard similarity

> 用 minhash 和 Jaccard 相似度去除近似重复

- Result: 3.1 TB of code

> 结果：3.1 TB 代码

Stack v2 [Lozhkov+ 2024]

- Issues, comments, PRs from GitHub Archive

> 来自 GitHub Archive 的 issues、评论、PR

- Repositories from the Software Heritage

> 仓库来自 Software Heritage

- Documentation from crawling websites (e.g., PyPI, npm, devdocs.io)

> 文档来自爬取的网站（如 PyPI、npm、devdocs.io）

- Processing: remove binary files, malware, bot activity, deduplication, PII redaction, subsample PRs

> 处理：移除二进制文件、恶意软件、机器人活动，去重、PII 脱敏、对 PR 采样

- Pair source code (especially low-resource languages like Nim) with shared low-level intermediate language (LLVM)

> 把源代码（尤其是 Nim 这类低资源语言）与共享的底层中间语言（LLVM）配对

- Include existing datasets (GSM8K, code contests, StackOverflow, arXiv, Wikipedia, OpenWebMath)

> 纳入已有数据集（GSM8K、代码竞赛、StackOverflow、arXiv、Wikipedia、OpenWebMath）

Pull requests:

> Pull requests：

- Linearize structured object to token sequence

> 将结构化对象线性化为 token 序列

- Add some inline context (e.g., file surrounding diff), subsample

> 加入一些行内上下文（如 diff 周边的文件内容），并做采样

```
<pr>Title: title\nusername_0: description
<pr_status>opened
<repo_name>reponame

<pr_base>
<pr_file>filepath_1
<pr_base_code>file_content/changes_1
...
<pr_file>filepath_N
<pr_base_code>file_content/changes_N

<pr_diff>
<pr_file>filepath_1
<pr_diff_hunk>diff_hunk_1
...
<pr_diff_hunk>diff_hunk_K
...
<pr_file>filepath_M
<pr_diff_hunk>diff_hunk_1
...
<pr_diff_hunk>diff_hunk_J
```

```
<pr_comment>username_id: comment
<pr_event_id>comment_id
...
...
...
<pr_review>username_id: review_comment\n
<pr_event_id>review_id
<pr_review_state>[approved, rejected, commented, changes_required]
...
...
...
<pr_review_comment>
<pr_event_id>comment_id
<pr_in_reply_to_review_id>review_id (opt)
<pr_in_reply_to_comment_id>comment_id (opt)
<pr_file>filepath
<pr_diff_hunk_comment_line>line_number
<pr_diff_hunk>diff_hunk_content
<pr_comment>username_id: comment
```

Recall:

> 回顾：

- Almost all data on the Internet is copyrighted.

> 互联网上几乎所有数据都受版权保护。

- Some of it is permissively licensed.

> 其中一部分采用宽松许可。

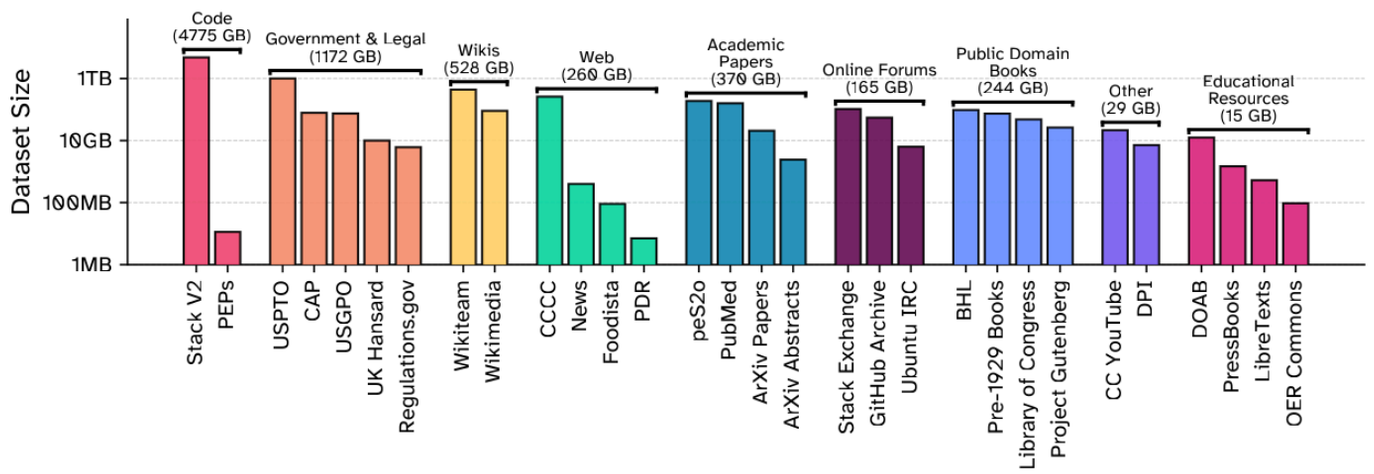
- Fair use of copyrighted content is not settled.

> 对版权内容的合理使用尚无定论。

Key question: can you train a good model using only permissively-licensed data?

> 关键问题：只用宽松许可的数据能训练出好模型吗？

CommonPile [Kandpal+ 2025]



- Collected 8TB dataset of permissively licensed data

> 收集了 8TB 的宽松许可数据集

Subtleties:

> 一些微妙之处：

- License laundering: redistribute copyrighted work under permissive license (hard to detect)

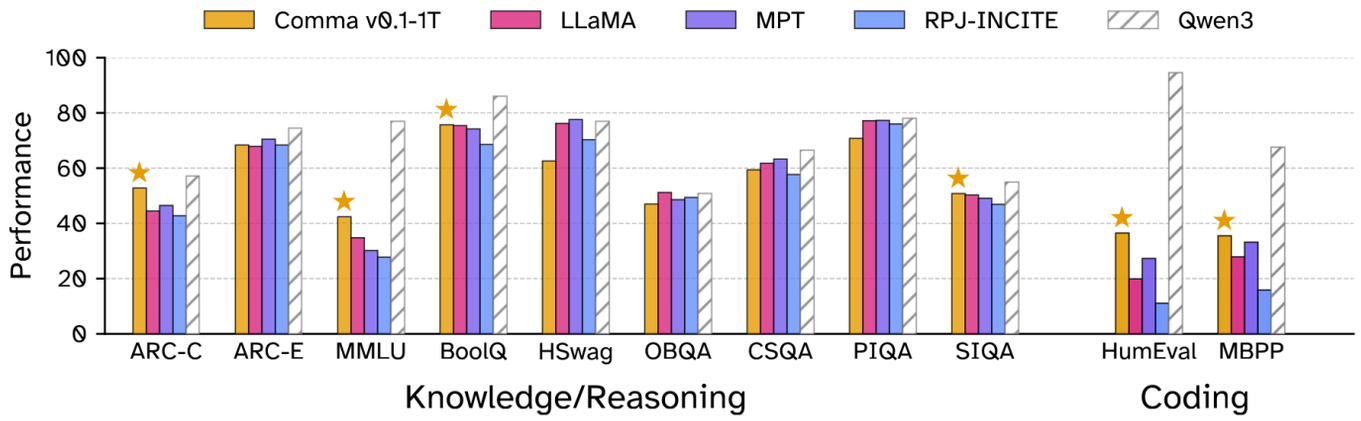
> 许可洗白：以宽松许可再分发受版权保护的作品（很难检测）

- Collection licenses (Dolma is ODC-By) doesn't extend to individual

> 数据集整体的许可（Dolma 是 ODC-By）并不延伸到其中的单个内容

- Synthetic data from LMs trained on unlicensed data is unclear

> 由在未授权数据上训练的语言模型生成的合成数据，法律地位不明



- Can do decently, but tough to compete without more tokens

> 能做得还不错，但缺更多 token 就很难有竞争力